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Saveetha Institute of Medical And Technical Sciences
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DESIGN OF A FASHION RETAIL TREND AND SEASONAL DEMAND VISUALIZATION PLATFORM

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

DSA0601- Data Handling and Visualization
to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY
IN

Artificial intelligence and data science
Submitted by

M.Arshini (192424322)

Under the Supervision of

Dr. S Senthilvadivu

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

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Chennai-602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Project Title” has been carried out by M.Arshini(192424322)under the supervision of “DR.S.Senthilvadivu”and is submitted in partial fulfilment of the requirements for the current semester of the “B.Tech ARTIFICIAL INTELLIGENCE AND DATA SCIENCE” program at Saveetha Institute of Medical and Technical Sciences, Chennai.

**Course Coordinator
Dr Sriramya P
Program Director
Department of CSE
Saveetha School of Engineering**

**Course Faculty
Dr. S Senthilvadivu
Professor
Department of CSE
Saveetha School of Engineering**

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Saveetha Institute of Medical and Technical Sciences

Chennai-602105

DECLARATION

We, [M.Arshini(192424322)] of the ARTIFICIAL INTELLIGENCE AND DATA SCIENCE, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “[“Design of a Fashion Retail Trend and Seasonal Demand Visualization Platform,”]” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place:chennai

Date:

**Signature of the Students
M.Arshini**

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Saveetha Institute of Medical and Technical Sciences

Chennai-602105

Individual Contribution Statement

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Student 1	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[__%]
Student 2	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[__%]
Student 3	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[__%]
Student 4	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[__%]
Total					100%

ABSTRACT

The project titled “Design of a Fashion Retail Trend and Seasonal Demand Visualization Platform” focuses on analyzing fashion retail sales data to identify seasonal demand patterns, popular product categories, styles, and relationships between different sales factors. Fashion retail generates large amounts of data from products, categories, styles, seasons, sales volume, prices, and customer preferences. However, raw sales data is often difficult to understand directly and may not provide clear information for effective inventory planning and merchandising decisions. Therefore, an interactive data visualization platform is proposed to transform complex sales data into meaningful and easy-to-understand insights.

The proposed platform uses fashion sales data that is cleaned, organized, and categorized before performing analysis. Power Query is used for data cleaning and transformation, while Power BI is used to create interactive visualizations and dashboards. The analysis considers different fashion categories, styles, seasons, sales volumes, prices, and popularity scores. Seasonal sales are analyzed across Spring, Summer, Autumn, and Winter to understand changes and variations in customer demand. Violin Plots are used to visualize the distribution of sales volume across different seasons, helping identify variations and demand patterns. Sunburst Charts are used to represent the hierarchical relationship between categories and styles and their contribution to sales.

The platform also analyzes monthly sales trends using Time-Series visualizations and applies the Moving Average technique to smooth short-term fluctuations and identify overall demand patterns. In addition, a Bubble Chart is used to study the relationship between product price, popularity, and sales volume. Interactive dashboard features such as season, month, and style slicers allow users to filter and explore the sales data according to their requirements.

The proposed platform converts large and complex fashion sales data into visual information that can be interpreted more easily. The analysis can help identify high-demand seasons, important product categories, popular styles, monthly demand patterns, and price-popularity relationships. These insights can support retailers in making better decisions related to inventory management, seasonal stock preparation, product selection, and merchandising.

Overall, the project provides an interactive and practical approach for understanding fashion retail trends and seasonal demand. Future improvements can include Artificial Intelligence and Machine Learning based demand forecasting, real-time sales monitoring, personalized product recommendations, cloud-based data storage, and mobile application support. The platform therefore provides a useful foundation for data-driven decision-making in the fashion retail industry.

Keywords: Fashion Retail, Seasonal Demand, Data Visualization, Power BI, Trend Analysis, Moving Average.

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CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

The fashion retail industry generates a large amount of sales data through physical stores and online platforms. This data contains information about product categories, styles, sales volume, prices, seasons, and monthly demand. However, raw sales data is often difficult to understand and use directly for decision-making.

The project “Design of a Fashion Retail Trend and Seasonal Demand Visualization Platform” focuses on converting raw fashion sales data into meaningful visual information. The proposed platform uses Microsoft Excel/CSV, Power Query, and Power BI for data cleaning, transformation, analysis, and visualization.

The platform analyzes fashion sales according to Spring, Summer, Autumn, and Winter seasons. Different visualization techniques such as violin plots, proportion charts, sunburst charts, time-series graphs, moving averages, and bubble charts are used to understand sales distribution, category contribution, style popularity, monthly demand, and price-popularity relationships.

This approach helps transform complex sales data into simple and interactive dashboards that can support better fashion retail planning and decision-making.

1.2 Need for the Project

Fashion demand changes according to seasons, customer preferences, product styles, and prices.

Retailers need to understand these changes to make better decisions about inventory and merchandising.

The project is needed to:

- Identify seasonal sales patterns.
- Analyze popular product categories and styles.
- Understand monthly demand trends.
- Study the relationship between price and product popularity.
- Present complex sales data through easy-to-understand visualizations.
- Support inventory and merchandising decisions.
- Help identify future demand trends using moving-average analysis.

1.3 Problem Statement

Fashion retailers generate large volumes of sales data, but raw data alone does not clearly show seasonal demand, popular categories, styles, or sales trends. Manual analysis can be time-consuming and may make it difficult to identify important patterns.

Therefore, there is a need to design a visualization platform that cleans, organizes, analyzes, and visually represents fashion retail sales data so that seasonal patterns, category and style

proportions, monthly trends, and price-popularity relationships can be easily understood.

1.4 Aim

The aim of this project is to design and develop a fashion retail trend and seasonal demand visualization platform that converts sales data into meaningful visual insights for better analysis and decision-making.

1.5 Objectives

The main objectives are:

1. To collect and organize fashion retail sales data.
2. To clean and transform the dataset using Power Query.
3. To classify sales data based on different seasons.
4. To analyze category and style proportions.
5. To visualize seasonal sales distribution using violin plots.
6. To identify monthly demand and sales trends.
7. To apply moving-average analysis for trend understanding.
8. To analyze the relationship between product price and popularity.
9. To develop an interactive dashboard using Power BI.
10. To support inventory and merchandising decisions through data visualization.

1.6 Scope

The scope of the project includes:

- Fashion retail sales data analysis.
- Seasonal demand analysis for Spring, Summer, Autumn, and Winter.
- Category and style-level analysis.
- Monthly trend analysis.
- Price versus popularity analysis.
- Interactive Power BI visualizations and slicers.
- Moving-average-based trend analysis.
- Supporting retail inventory and merchandising decisions.

Future versions can be enhanced using AI/ML forecasting, real-time dashboards, personalized recommendations, cloud databases, and mobile applications.

1.7 Expected Engineering Outcome

The expected outcome is an interactive fashion retail visualization platform that converts raw sales data into clear and useful insights. The platform should make it easier to identify seasonal demand patterns, popular categories and styles, monthly sales trends, and price-popularity relationships.

The final dashboard is expected to provide a simple and effective way for users to explore fashion sales data using interactive visualizations and slicers, supporting improved inventory planning, product planning, sales analysis, and merchandising decisions.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review focuses on existing approaches used for **fashion retail sales analysis, seasonal demand analysis, trend analysis, and data visualization**. The review considers how sales data is cleaned, transformed, analyzed, and presented using visualization tools.

The main areas considered are:

- Fashion retail sales analysis
- Seasonal demand analysis
- Category and style analysis
- Time-series trend analysis
- Moving-average analysis
- Interactive dashboard development
- Data visualization for retail decision-making

The reviewed approaches provide a foundation for developing the proposed **Fashion Retail Trend and Seasonal Demand Visualization Platform**.

2.2 Review of Existing Work

2.2.1 Fashion Retail Sales Analysis

Existing retail analysis approaches use historical sales data to understand product performance and customer demand. Sales data can contain information such as product category, style, price, sales volume, season, and month.

Analysis of these factors helps retailers identify products that perform well and understand changes in demand.

2.2.2 Seasonal Demand Analysis

Seasonal analysis is important in fashion retail because customer demand can change between **Spring, Summer, Autumn, and Winter**. Comparing sales across seasons helps identify periods of high and low demand.

Visual methods can make seasonal differences easier to understand than examining raw numerical data.

2.2.3 Data Visualization and Dashboards

Business intelligence tools such as **Power BI** are commonly used to convert processed data into interactive dashboards. Charts, graphs, slicers, and filters allow users to explore data from different perspectives.

For the proposed project, Power BI is used to represent seasonal sales, category proportions, style proportions, monthly trends, and price-popularity relationships.

team 8_DHV-Final ppt

2.2.4 Time-Series and Moving-Average Analysis

Historical monthly sales data can be represented using time-series graphs to identify increasing or decreasing trends. Moving-average analysis can be used to smooth short-term fluctuations and make the underlying demand trend easier to observe.

The proposed project applies these techniques to understand monthly fashion demand and support future planning.

team 8_DHV-Final ppt

2.2.5 Hierarchical Visualization

Sunburst charts can represent hierarchical relationships between different levels of data. In this project, the hierarchy is represented as:

Category → Style → Sales Volume

This provides a compact view of how individual styles contribute to different product categories.

2.3 Comparative Analysis:

Existing Approach	Main Purpose	Limitation
Raw sales tables	Store sales information	Difficult to identify patterns
Basic charts	Display sales values	Limited interactive analysis
Seasonal analysis	Compare seasonal demand	May not show category-style hierarchy
Time-series graphs	Analyze monthly trends	Does not provide complete product analysis
Moving average	Smooth demand trends	Focuses mainly on trend behavior
Power BI dashboards	Interactive visualization	Requires proper data preparation
Proposed platform	Combined fashion retail analysis	Designed specifically for seasonal, category, style and trend analysis

The proposed platform combines multiple analysis methods instead of depending on a single visualization. It integrates seasonal analysis, category and style proportions, time-series trends, moving averages, and price-popularity analysis in an interactive dashboard.

2.4 Research/Engineering Gap

From the existing approaches, a gap can be identified in presenting different fashion retail factors together in a simple and interactive platform.

Individual methods can analyze sales trends, seasonal demand, or product performance, but a combined visualization approach can provide a more complete view of fashion retail data.

The proposed project addresses this gap by integrating:

- Seasonal demand analysis
- Category proportion analysis
- Style proportion analysis
- Category-to-style hierarchy
- Monthly demand trends
- Moving-average analysis
- Price versus popularity analysis
- Interactive slicers and dashboards

This integration helps users examine fashion retail data from multiple perspectives within one

platform.

2.5 Proposed Contribution

The proposed project contributes an **interactive fashion retail trend and seasonal demand visualization platform** using Power Query and Power BI.

The major contributions are:

1. Cleaning and transforming raw fashion sales data.
2. Organizing sales data according to seasons.
3. Visualizing seasonal sales distribution using violin plots.
4. Showing category and style contributions.
5. Representing category-style relationships using a sunburst chart.
6. Analyzing monthly sales trends using time-series graphs.
7. Applying moving averages to understand demand trends.
8. Studying price and product popularity relationships using bubble charts.
9. Providing interactive filtering using Power BI slicers.
10. Converting complex sales information into visual insights that can support inventory, merchandising, and product planning decisions.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

The proposed **Fashion Retail Trend and Seasonal Demand Visualization Platform** requires the following functional and technical requirements.

3.1.1 Functional Requirements

1. The system should accept fashion retail sales data from Excel/CSV files.
2. The data should be cleaned and transformed using Power Query.
3. The system should classify sales according to Spring, Summer, Autumn, and Winter.
4. The system should analyze sales by product category and style.
5. The system should display seasonal sales distribution using violin plots.
6. The system should display category and style proportions.
7. The system should represent the Category → Style → Sales Volume relationship.
8. The system should analyze monthly demand trends.
9. The system should support moving-average analysis.
10. The system should analyze price versus product popularity.
11. The dashboard should provide interactive slicers for filtering data.

3.1.2 Non-Functional Requirements

- **Usability:** Dashboard should be simple and easy to understand.
- **Performance:** Visualizations should load efficiently for the available dataset.
- **Accuracy:** Calculations and visual representations should correctly reflect the input data.
- **Maintainability:** Data transformation and dashboard components should be easy to update.
- **Scalability:** The design should allow future integration of larger datasets.

3.2 Constraints and Applicable Standards

Constraints

The major constraints considered in the project are:

- Availability and quality of fashion sales data.
- Missing or inconsistent values in the input dataset.
- Limited information about customer behavior.
- Dashboard performance depends on dataset size and system resources.
- Analysis is dependent on the accuracy of historical sales data.
- Moving-average analysis provides trend understanding but does not guarantee future demand.

Applicable Practices

The project follows consistent practices for:

- Data cleaning and transformation.
- Structured data organization.
- Consistent naming of categories and styles.
- Appropriate selection of visualization techniques.
- Protection of sensitive or confidential sales information.

3.3 Design Specifications

The proposed system follows the following design:

Raw Sales Data → Data Cleaning → Data Transformation → Data Analysis → Visualization → Interactive Dashboard → Retail Insights

Main Components:

Component	Purpose
Excel/CSV	Input sales data
Power Query	Data cleaning and transformation
Power BI	Dashboard and visualization
DAX	Calculations and measures
Violin Plot	Seasonal sales distribution
Sunburst Chart	Category and style hierarchy
Time-Series Graph	Monthly demand trends
Moving Average	Trend smoothing
Bubble Chart	Price-popularity analysis
Slicers	Interactive filtering

3.4 Alternative Solutions and Evaluation:

Solution	Advantages	Limitations
Microsoft excel	Easy to use and widely available	Limited dashboard interaction
Python visualization	Flexible and highly customizable	Requires programming knowledge
Tableau	Strong visualization capabilities	Additional software/platform dependency
Power bi	Interactive dashboards, slicers and business analytics	Requires proper data preparation
Proposed power bi platform	Combines seasonal category, style	Depends on quality of input data

Different approaches were considered for presenting and analyzing the fashion sales data. Based on the project requirements, **Power BI with Power Query** was selected because it supports data transformation, interactive visualization, filtering, and dashboard development.

3.5 Selected Design and Detailed Design

The selected design consists of five main stages.

Stage 1 – Data Collection

Fashion retail sales data is collected in Excel/CSV format. The dataset contains information required for sales, category, style, season, price, and time-based analysis.

Stage 2 – Data Cleaning

Power Query is used to clean and transform the dataset. Incorrect, missing, or inconsistent data is processed before analysis.

Stage 3 – Data Analysis

The cleaned data is analyzed based on:

- Season
- Category
- Style
- Sales volume
- Month
- Price
- Popularity

Stage 4 – Visualization

Different visualizations are selected according to the analysis requirement:

- Violin plots for seasonal sales distribution.
- Proportion charts for category and style contribution.
- Sunburst charts for category-style hierarchy.
- Time-series graphs for monthly demand.
- Moving averages for trend analysis.
- Bubble charts for price-popularity relationships.

Stage 5 – Interactive Dashboard

The visualizations are combined into a Power BI dashboard. Interactive slicers allow users to filter the information by season and style.

3.6 Trade-off Analysis

During the design process, different trade-offs were considered.

Design Decision	Benefit	Trade-off
Power BI instead of only Excel	Better interactive visualization	Requires Power BI knowledge
Multiple visualization types	Provides different analytical views	Dashboard can become complex
Moving average	Makes trends easier to observe	May hide short-term variations
Sunburst chart	Shows hierarchy clearly	Can become difficult to read with many categories
Interactive slicers	Allows user-based filtering	Requires proper dashboard design
Power Query transformation	Improves data consistency	Adds a preprocessing stage

3.7 Sustainability, Ethics, Safety, Risk and Security

3.7.1 Sustainability

The project supports sustainable decision-making by helping retailers understand demand patterns before planning inventory. Better demand analysis can help reduce unnecessary overstocking and improve product planning.

The digital dashboard also reduces dependence on manually prepared reports and supports efficient use of sales information.

3.7.2 Ethics

Ethical considerations include:

- Using reliable and accurate sales data.
- Avoiding misleading visualizations.
- Clearly presenting analytical results.
- Protecting confidential business information.
- Avoiding unauthorized use of customer-related information.
- Using data only for its intended analytical purpose.

3.7.3 Safety

The project is a software-based data visualization system and does not involve physical machinery or hazardous materials. Therefore, physical safety risks are minimal.

However, users should ensure that decisions are not made solely from automatically generated trends without appropriate business judgment.

3.7.4 Security

Security measures should be considered when handling retail sales information. Access to datasets and dashboards should be limited to authorized users. Sensitive business information should not be unnecessarily exposed.

Regular backups should also be maintained to prevent data loss. If future versions include cloud databases or real-time data, stronger authentication and access-control mechanisms should be implemented.

3.8 Risk Register

Risk ID	Risk	Probability	Impact	Mitigation
R1	Missing or incomplete data	Medium	High	Data validation and cleaning
R2	Incorrect data transformation	Medium	High	Verify Power Query steps
R3	Incorrect visualization	Low	Medium	Test each visual
R4	Dashboard performance issues	Medium	Medium	Optimize dataset and visuals
R5	Data security problems	Low	High	Restrict access and maintain backups
R6	Incorrect interpretation	Medium	Medium	Provide clear labels and explanations

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project is developed using a modular approach.

The fashion sales dataset is first processed using Pandas. The cleaned data is then analyzed using different visualization techniques.

Plotly is used to create interactive visualizations.

The implementation consists of three major modules:

Module 1: Fashion Sales Data Wrangling & Category Tagging

Module 2: Seasonal Demand Distribution & Proportion

Module 3: Trend Forecasting & Style Association

4.2 Implementation

Module 1: Fashion Sales Data Wrangling & Category Tagging

The fashion sales dataset is collected as the primary input for the project and loaded into Python using the Pandas library. The dataset is examined to understand its structure, columns, data types, and sales-related information. Basic data validation is performed to ensure that the data is suitable for further processing and analysis.

The data is cleaned by checking for missing values, duplicate records, incorrect entries, and inconsistent formats. Missing or invalid values are handled appropriately, and duplicate records are removed to improve data quality. Numerical and categorical fields are also standardized so that the dataset remains consistent and reliable.

After cleaning, the products are organized into meaningful categories based on product category, style, and season. These tags help in identifying sales patterns and comparing different fashion segments. The final cleaned and categorized dataset is then prepared for **data analysis, visualization, and dashboard development** in the subsequent modules.

Module 2: Seasonal Demand Distribution & Proportion

- Sales are grouped according to season.
- Seasonal sales distribution is analyzed.
- Violin Plot is created to compare seasonal sales.
- Category and style information is grouped hierarchically.
- Sunburst Chart is created to display category-style proportion.

Module 3: Trend Forecasting & Style Association

- Monthly sales data is analyzed.
- Multiple Time-Series graphs are created.
- Moving Average is calculated to smooth sales fluctuations.
- Overall demand trends are identified.
- Price and popularity are analyzed.
- Bubble Chart is created using price, popularity, and sales volume.

Requirement	Test Method	Acceptance Criterion	Status
Dataset Input	Load dataset	Dataset loads successfully	To Test
Data Cleaning	Check missing/duplicate values	Clean dataset generated	To Test
Category Tagging	Check categories	Categories correctly assigned	To Test
Violin Plot	Generate seasonal plot	Seasonal distribution displayed	To Test
Sunburst Chart	Generate chart	Category-style hierarchy displayed	To Test
Time-Series	Generate monthly graph	Monthly trends displayed	To Test
Moving Average	Calculate average	Trend line displayed	To Test
Bubble Chart	Generate chart	Price-popularity relationship displayed	To Test
Interactive Output	Test Plotly controls	Visualization responds correctly	To Test

4.4 Results

Figure 4.2 – Seasonal Sales Violin Plot

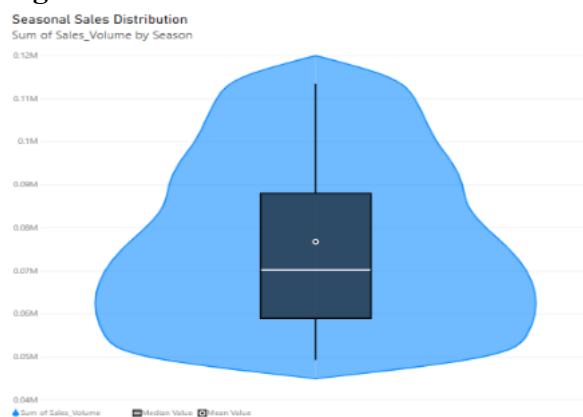


Figure 4.3 – Category-Style Sunburst Chart

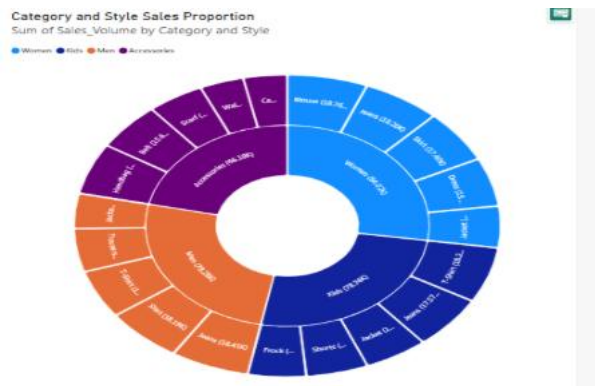
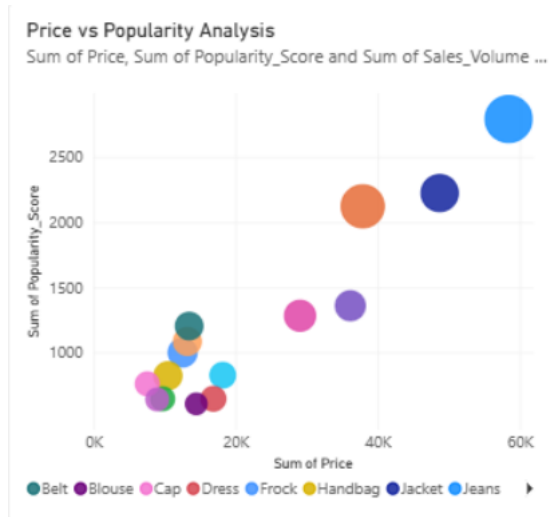


Figure 4.4 – Monthly Sales Time-Series & Moving Average Trend



Figure 4.5 – Price-Popularity Bubble Chart



4.5 Data Analysis & Engineering Judgment

The platform is evaluated based on the quality of data processing and the clarity of the generated visualizations.

The Violin Plot is used to understand seasonal sales distribution. The Sunburst Chart provides a hierarchical representation of categories and styles.

The Time-Series visualization is used to identify monthly demand patterns, while Moving Average analysis helps reduce short-term fluctuations and identify the overall trend.

The Bubble Chart helps analyze price, popularity, and sales volume together.

The final visualizations are interpreted based on the actual dataset and results obtained during implementation.

4.6 Achievement of Objectives

Objective	Evidence	Achievement
Clean sales data	Processed dataset	To be evaluated
Categorize products	Category/style/season fields	To be evaluated
Analyze seasonal demand	Violin Plot	To be evaluated
Analyze category-style proportion	Sunburst Chart	To be evaluated
Analyze monthly trends	Time-Series	To be evaluated
Identify demand trends	Moving Average	To be evaluated
Analyze price-popularity	Bubble Chart	To be evaluated
Create interactive platform	Plotly output	To be evaluated

4.7 Strengths & Limitations

Strengths

- Easy-to-understand visualizations.
- Interactive Plotly-based platform.
- Multiple visualization techniques are integrated.
- Helps identify seasonal demand patterns.
- Supports category and style analysis.
- Shows monthly demand trends.
- Helps analyze price and popularity relationships.

Limitations

- Results depend on the quality of the dataset.
- Moving Average provides a simple trend estimation.
- Missing data may affect analysis.
- The platform depends on the available sales parameters.
- It does not currently use advanced Machine Learning forecasting

4.8 Project Planning

Phase	Activity
1	Literature Review
2	Dataset Collection
3	Data Cleaning
4	Category Tagging
5	Seasonal Visualization
6	Sunburst Visualization
7	Time-Series Analysis
8	Moving Average
9	Bubble Chart
10	Testing
11	Results

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project “Design of a Fashion Retail Trend and Seasonal Demand Visualization Platform” was developed to analyze fashion retail sales data and present useful information through interactive visualizations.

The project successfully used Power Query for data cleaning and transformation and Power BI for data analysis and visualization. The dashboard provides information about seasonal demand, product categories, styles, monthly sales trends, moving averages, and price-popularity relationships.

The developed platform makes complex sales data easier to understand and supports better inventory planning, merchandising, sales analysis, and product planning. The project demonstrates how data visualization can convert raw retail data into meaningful business insights.

5.2 Future Work

The project can be improved in the future by adding:

1. AI/ML Forecasting – Predict future fashion demand.
2. Real-Time Dashboard – Display continuously updated sales information.
3. Personalized Recommendations – Recommend products based on customer preferences.
4. Cloud Database – Store and manage large volumes of sales data.
5. Mobile Application – Provide dashboard access through mobile devices.
6. Advanced Analytics – Include more detailed customer and product analysis.

Professional Learning & Individual Reflection

New Knowledge Acquired:

- Learned how to clean and transform fashion retail sales data using Power Query.
- Learned to categorize sales data based on season, category, and style.
- Gained knowledge of creating interactive visualizations using Power BI.
- Learned about seasonal demand analysis, sales distribution, and category/style proportions.
- Understood time-series analysis and moving average techniques for identifying demand trends.
- Learned how dashboards can support retail inventory and merchandising decisions.

Engineering & Professional Skills Developed:

- Data cleaning, preprocessing, and transformation skills.
- Data visualization and dashboard development using Power BI.
- Analytical and problem-solving skills.
- Time-series and trend analysis skills.

- Teamwork, task coordination, and project management.
- Technical documentation and presentation skills.

Individual Reflection

•M.Arshini

The most difficult problem was selecting suitable visualizations to clearly represent seasonal demand and fashion trends. I solved it by using Power BI charts such as violin plots, sunburst charts, time-series graphs, and bubble charts. I decided to use interactive slicers because they allow users to filter the dashboard by season and style. I independently improved my knowledge of Power BI and DAX. If repeated, I would design a more advanced and user-friendly dashboard.

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APPENDICES

Appendix A – Implementation

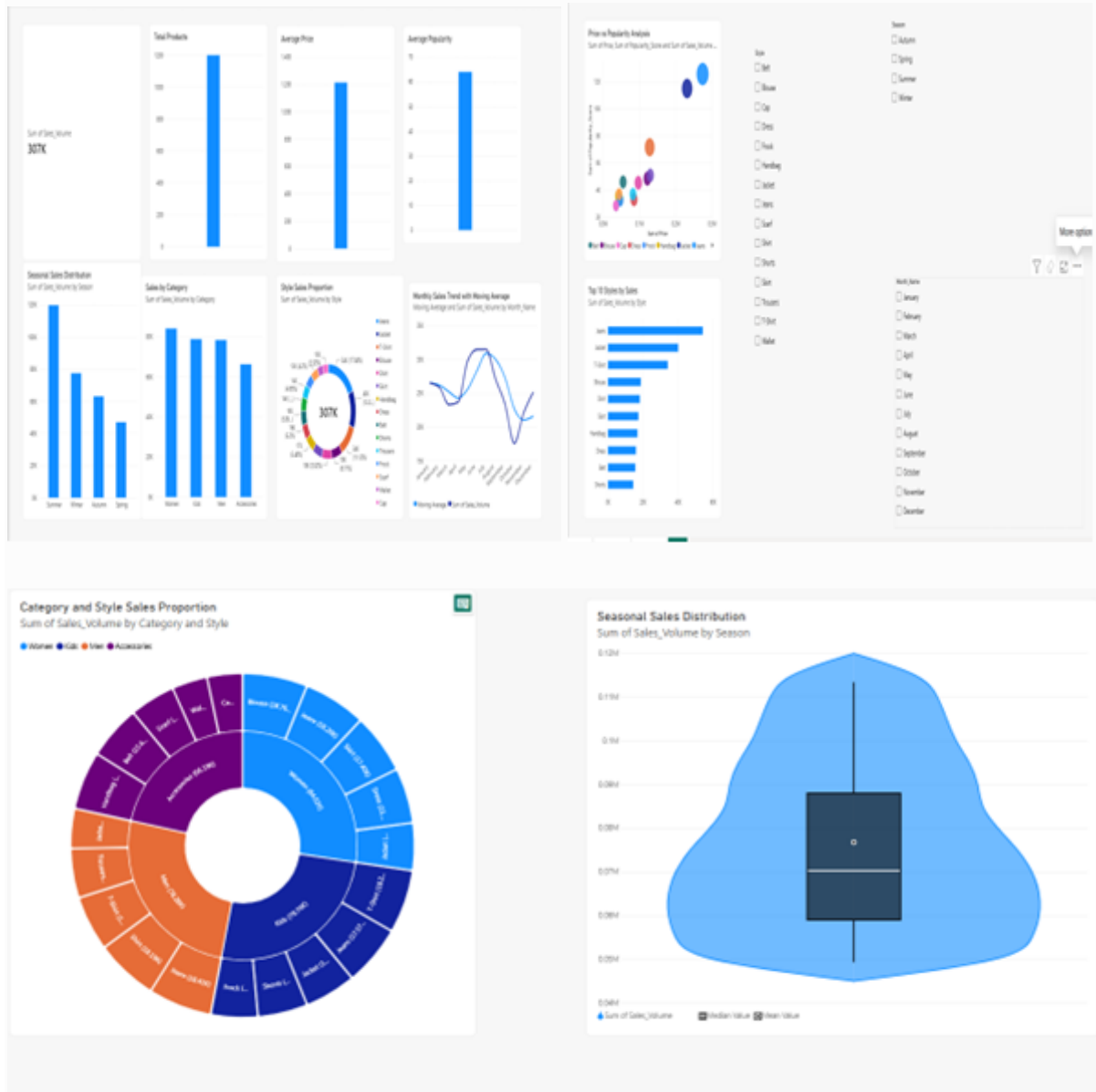
The project was developed using Microsoft Power BI and Power Query. The main implementation activities included:

- Dataset import and loading
- Data cleaning and transformation
- Missing-value and duplicate handling
- Category, style, and season classification
- Creation of required visualizations
- Time-series and moving-average analysis
- Interactive dashboard development using filters and slicers

Appendix B – Dataset Details

Parameter	Details
Input Type	Fashion Retail Sales Dataset
File Format	CSV
Data Processing	Power Query
Visualization Tool	Power BI
Category	Fashion product category
Style	Product style
Season	Spring / Summer / Autumn / Winter
Sales Volume	Number of products sold
Price	Product price
Popularity Score	Product popularity
Month	Month of sale

Appendix C – Output



Appendix D – Test Results

Test Case	Actual Result	Status
Dataset Loading	Dataset loaded successfully	Pass
Data Cleaning	Data cleaned using Power Query	Pass
Category Tagging	Categories assigned successfully	Pass
Visualizations	Required charts displayed correctly	Pass
Dashboard	Interactive dashboard worked successfully	Pass

Appendix E – Performance Results

Parameter	Result
Dataset Records	Actual dataset records
Categories	Actual number of categories
Seasons	Actual number of seasons
Visualization Types	5
Processing Tool	Power Query
Visualization Tool	Power BI
Trend Analysis	Moving Average

Appendix F – Software and Hardware Requirements

Software Requirements

- Microsoft Power BI Desktop
- Power Query
- Microsoft Windows
- Web Browser

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB minimum, 8 GB recommended
- Storage: 10 GB available space
- System: Standard laptop/desktop
- Internet: Required for dataset collection when needed